Intro to Machine Learning

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Content

- What is Machine Learning?
- Supervised Learning
- Unsupervised Learning
- Linear Regression with One Variable
- Linear Regression with Multiple Variables

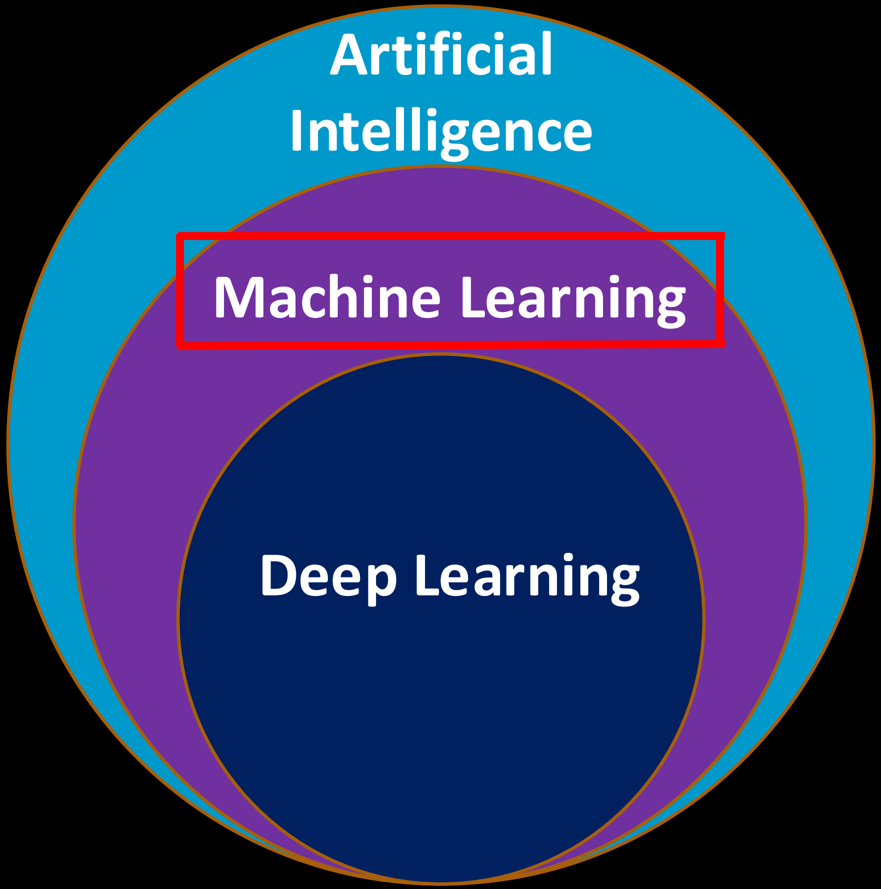
What is Machine Learning?

Machine Learning (ML) – the use and development of computer systems that are able to learn and adapt without following explicit instructions, by using algorithms and statistical models to analyse and draw inferences from patterns in data.

Machine Learning (ML) – Field of study that gives computers the ability to learn without being explicitly programmed. Arthur Samuel (1959).

What is Machine Learning?

Machine Learning is a part of Artificial Intelligence.



What is Machine Learning?

Machine learning

- Supervised learning
- Unsupervised learning
- Reinforcement learning
- Recommender system

Content

- ~~What is Machine Learning?~~
- Supervised Learning
- Unsupervised Learning
- Linear Regression with One Variable
- Linear Regression with Multiple Variables

Supervised Learning

Supervised learning

A. Classification problem

- Logistic Regression
- Decision Tree
- Naive Bayes
- K – Nearest Neighbor
- Support Vector Machine

Example:

- Email spam detection
- Speech Recognition

B. Regression problem

- **Linear Regression**
- **Ridge Regression**
- **Stepwise Regression**

Example:

- **Stock Market Prediction**
- **Rainfall prediction**

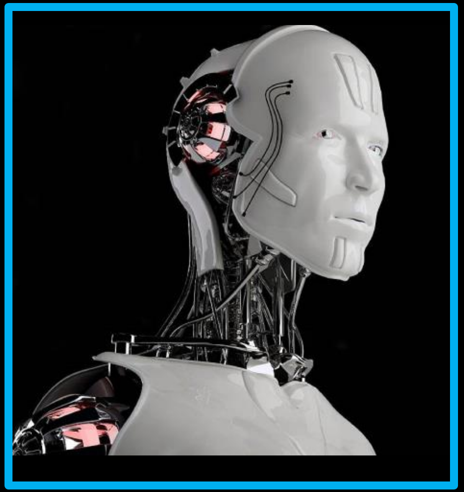
Supervised Learning

Classification problem

Apple

Banana

Input



Model
(Algorithm)

Apple

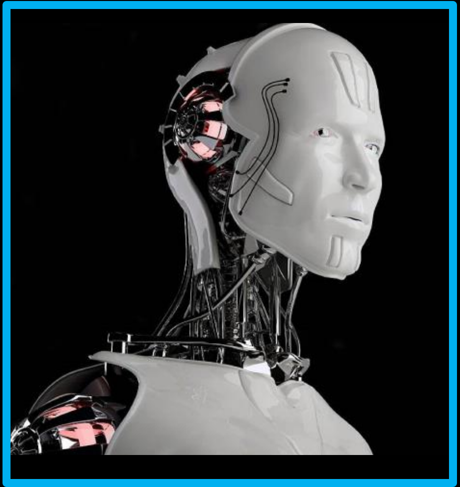
Output

Supervised Learning

Regression problem

45 °C

Temperature (°C)	Temperature (F)
10	50
13	55.4
22	71.6
35	95



113 F

Input

Model (Algorithm)

Output

Content

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Unsupervised Learning

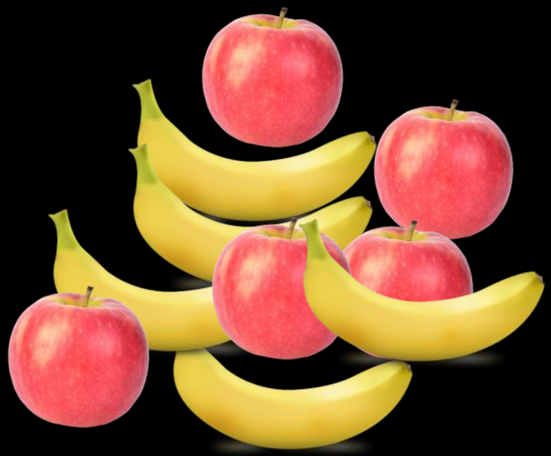
Unsupervised learning

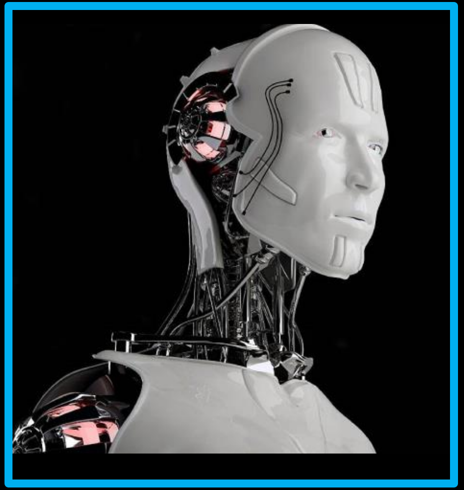
Clustering problem
- K – Means
- MeanShift

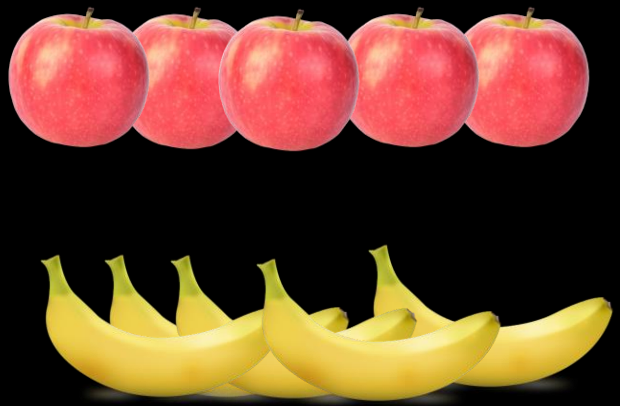
Dimensionality reduction
- Principle Component Analysis (PCA)
- Linear Discriminant Analysis (LDA)
- Autoencoders (AEs)

Unsupervised Learning

Clustering problem







Input

Model
(Algorithm)

Output

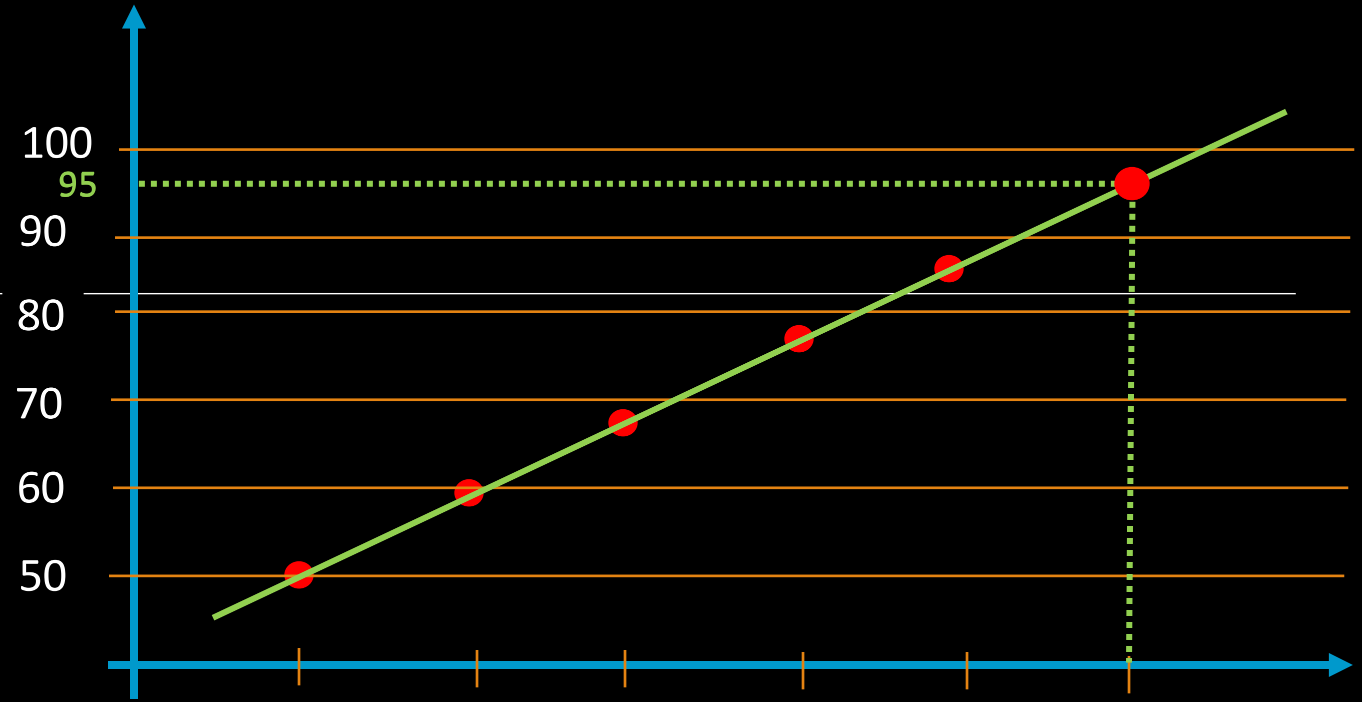
Content

- ~~What is Machine Learning?~~
- ~~Supervised Learning~~
- ~~Unsupervised Learning~~
- Linear Regression with One Variable
- Linear Regression with Multiple Variables

Dataset

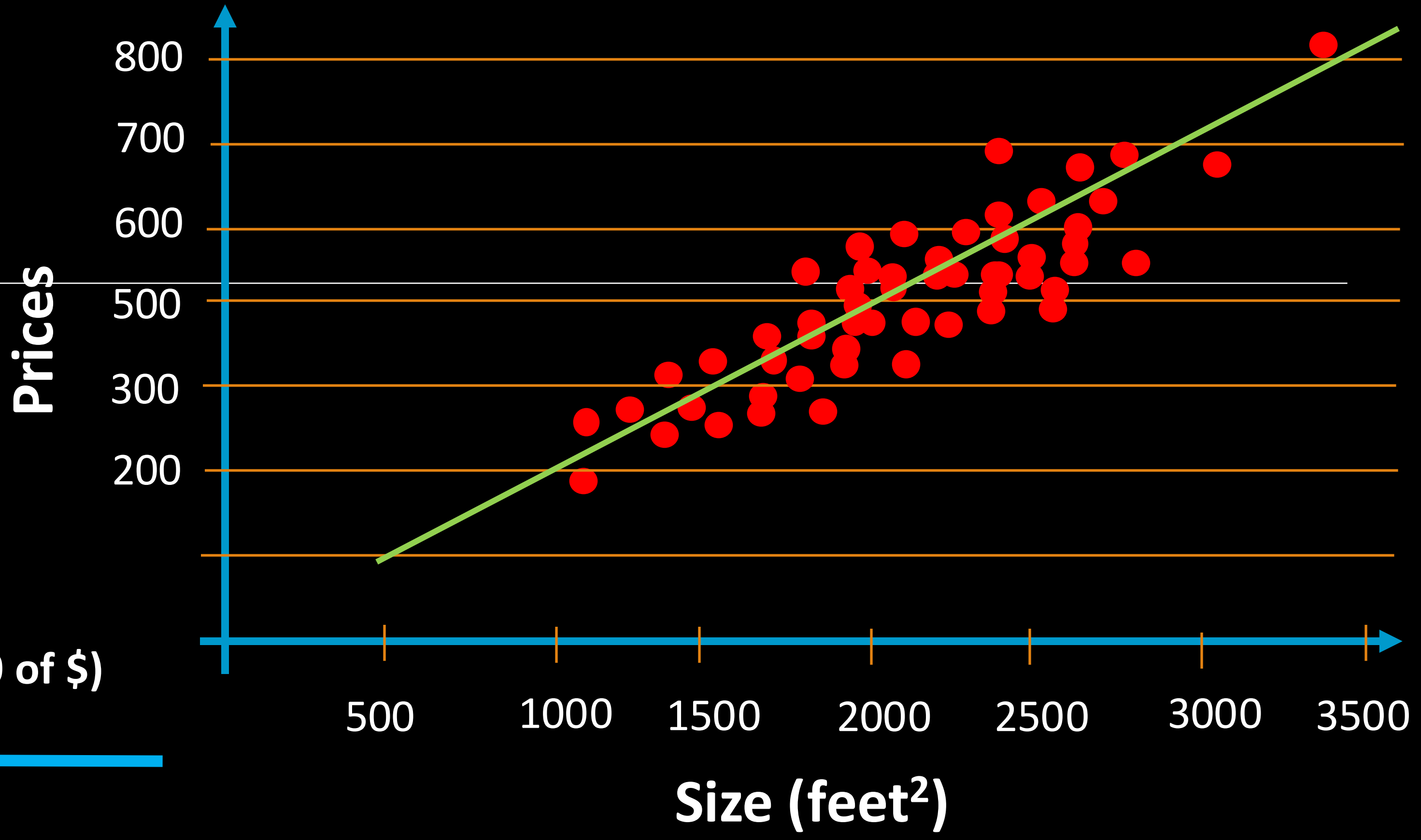
Temp (°C) (x)	Temp (F) (y)
10	50
15	59
20	68
25	77
30	86
35	

$$F = 32 + 1.8 \cdot t$$



Houses prices
(Dataset)

Size ($feet^2$), (x)	Prices (in 1000 of $) ($y$)
2104	460
1416	232
1534	315
852	178
…	…



Training set of Temp. Dataset

m	Temp (C°), (x)	Temp F (y)
1	10	50
2	15	59
3	20	68
4	25	77
…	…	…

x = "input" variable/feature

y = "output" variable/target

m = Number of training examples

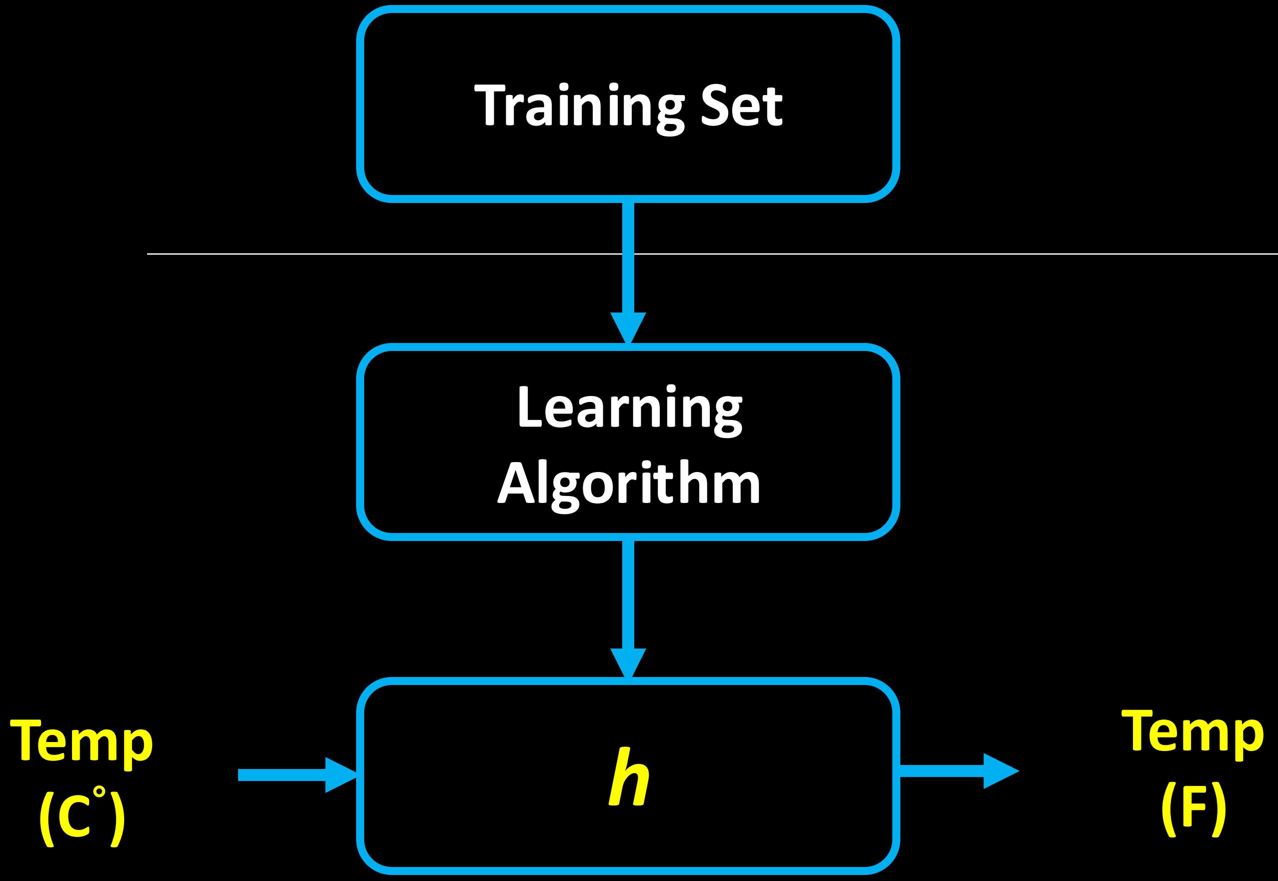
(x, y) – one training example

$(x^{(i)}, y^{(i)})$ – i th training example

$(x^{(1)}, y^{(1)}) = (\quad , \quad)$

$(x^{(2)}, y^{(2)}) = (\quad , \quad)$

$(x^{(3)}, y^{(3)}) = (\quad , \quad)$



Hypothesis

$y = h_\theta(x) =$

$\theta_0 = b =$

$\theta_1 = w =$

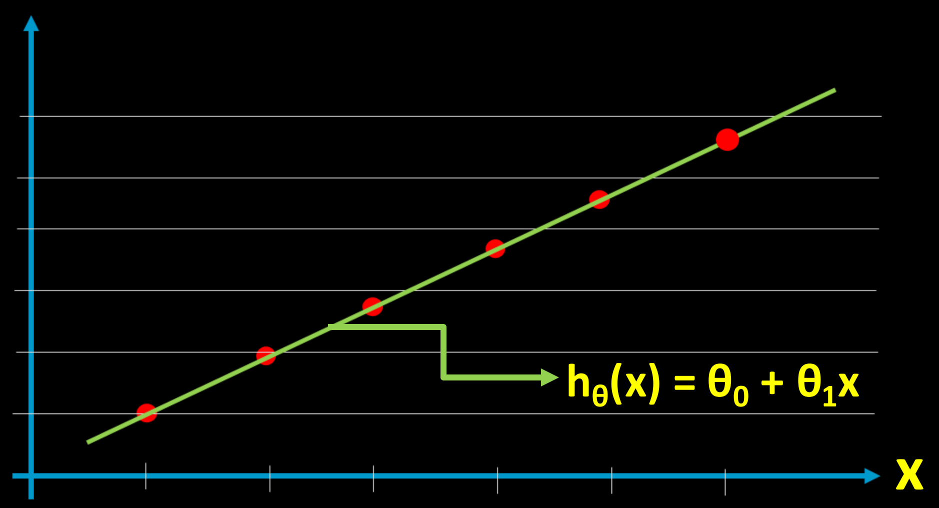
$x =$

How do we represent h?

$h_\theta(x) = \theta_0 + \theta_1 x$

$y = b + wx$

y



$F = 32 + 1.8 \cdot t$

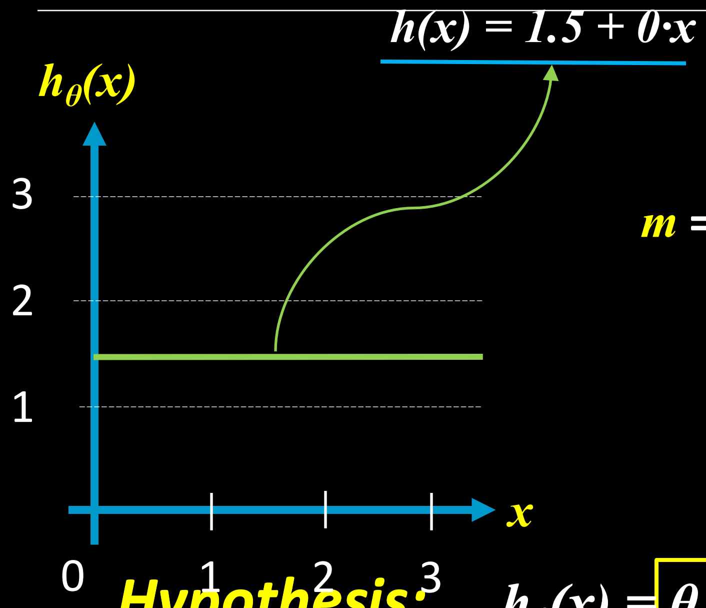
Linear Regression with one variable

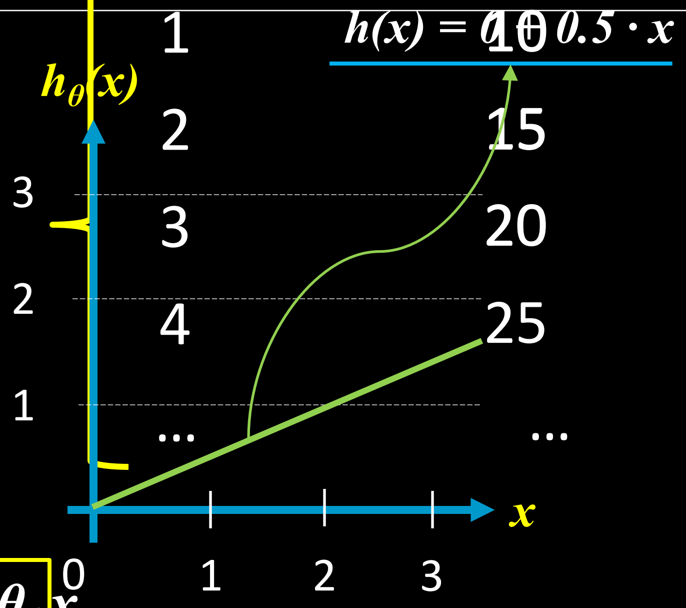
Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$

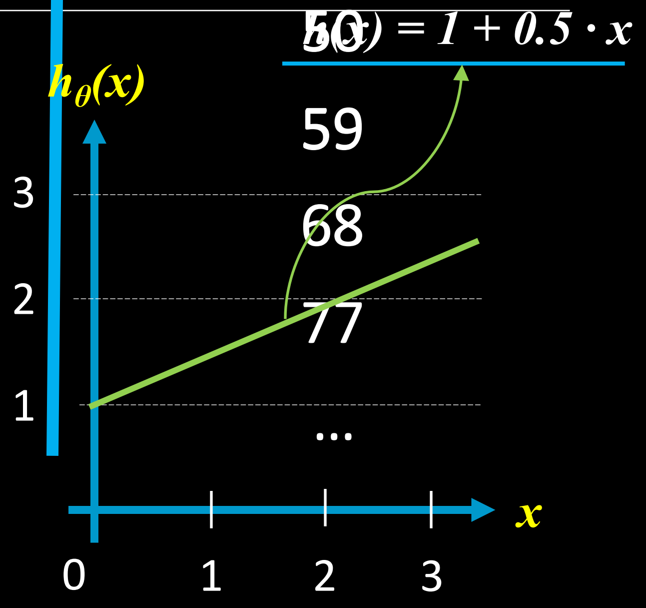
Temperature Dataset

$m = 50$

	Temp (C°), (x)	Temp F (y)
1	10	50
2	15	59
3	20	68
4	25	77
…	…	…







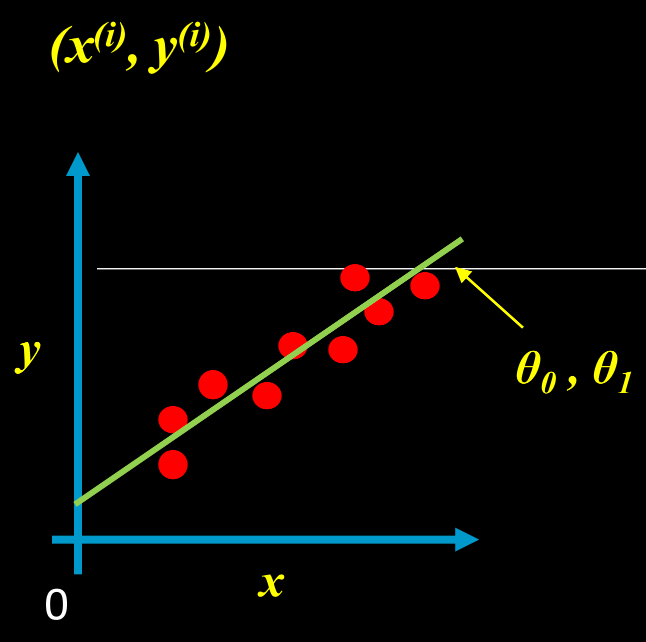
Hypothesis: $h_\theta(x) = \boxed{\theta_0} + \boxed{\theta_1} x$

θ_i : Parameters

$\theta_0 = 1.5$
$\theta_1 = 0$

$\theta_0 = 0$
$\theta_1 = 0.5$

$\theta_0 = 1$
$\theta_1 = 0.5$



Idea: Choose θ_0, θ_1 so that $h_\theta(x)$ is close to y for our training examples (x, y).

$$\underset{\theta_0,\ \theta_1}{\text{minimize}} \ \frac{1}{2m} \sum_{i=1}^{m} \Big(\underbrace{h_\theta(x^{(i)})}_{h_\theta(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}} - y^{(i)} \Big)^2$$

$$\boxed{J(\theta_0, \theta_1) = } \rightarrow \text{Cost function}$$

Squared error function

Hypothesis:

$$h_\theta(x) = \theta_0 + \theta_1 x$$

Parameters:

$$\theta_0 , \theta_1$$

Cost function:

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

Goal:

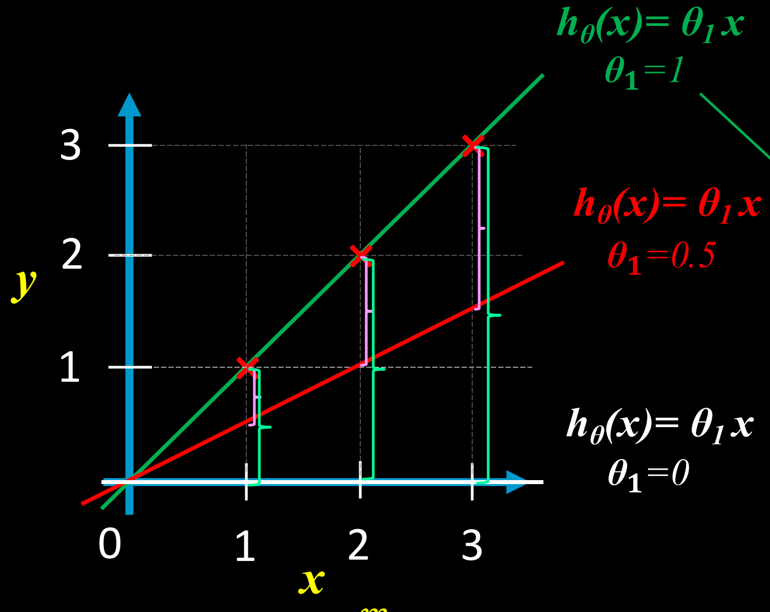
$$\underset{\theta_0, \theta_1}{\text{minimize}} \quad J(\theta_0, \theta_1)$$

Simplified:

$$= 0$$

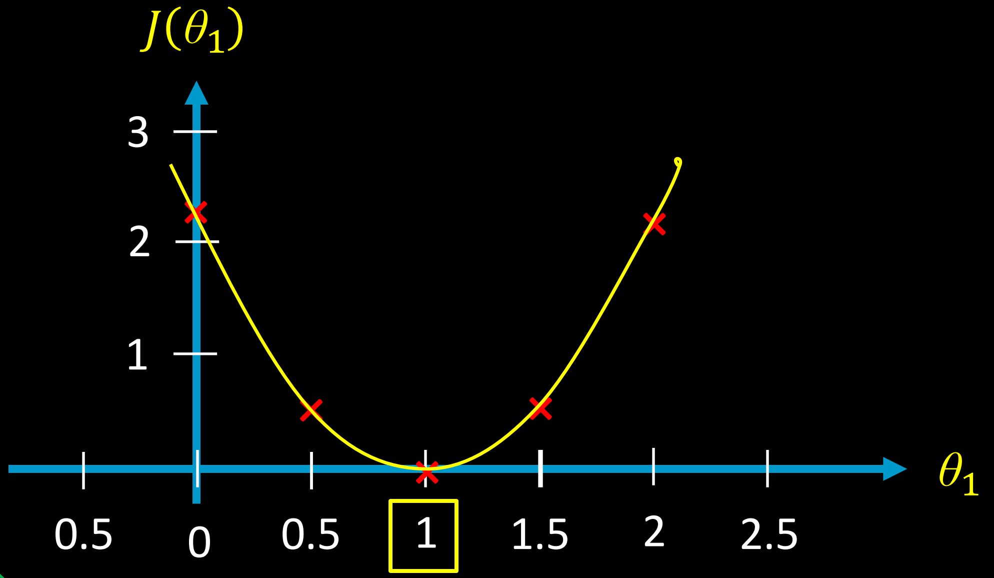
$$J(\theta_1) =$$

$h_\theta(x)$



$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

$J(\theta_1)$



$$J(1) = \frac{1}{2m}\left((1-1)^2 + (2-2)^2 + (3-3)^2\right) = \frac{1}{2\cdot 3}\cdot 0 = 0$$

$$J(0.5) = \frac{1}{2m}\left((0.5-1)^2 + (1-2)^2 + (1.5-3)^2\right) = \frac{1}{2\cdot 3}\cdot 3.5 = 0.58$$

$$J(0) = \frac{1}{2m}\left((0-1)^2 + (0-2)^2 + (0-3)^2\right) = \frac{1}{2\cdot 3}\cdot 14 = 2.33$$

θ_1	1	0.5	0	1.5	2
$J(\theta_1)$	0	0.58	2.33	0.58	2.33

$$\underset{\theta_1}{\text{minimize}}\ J(\theta_1)$$

Cost function: $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$

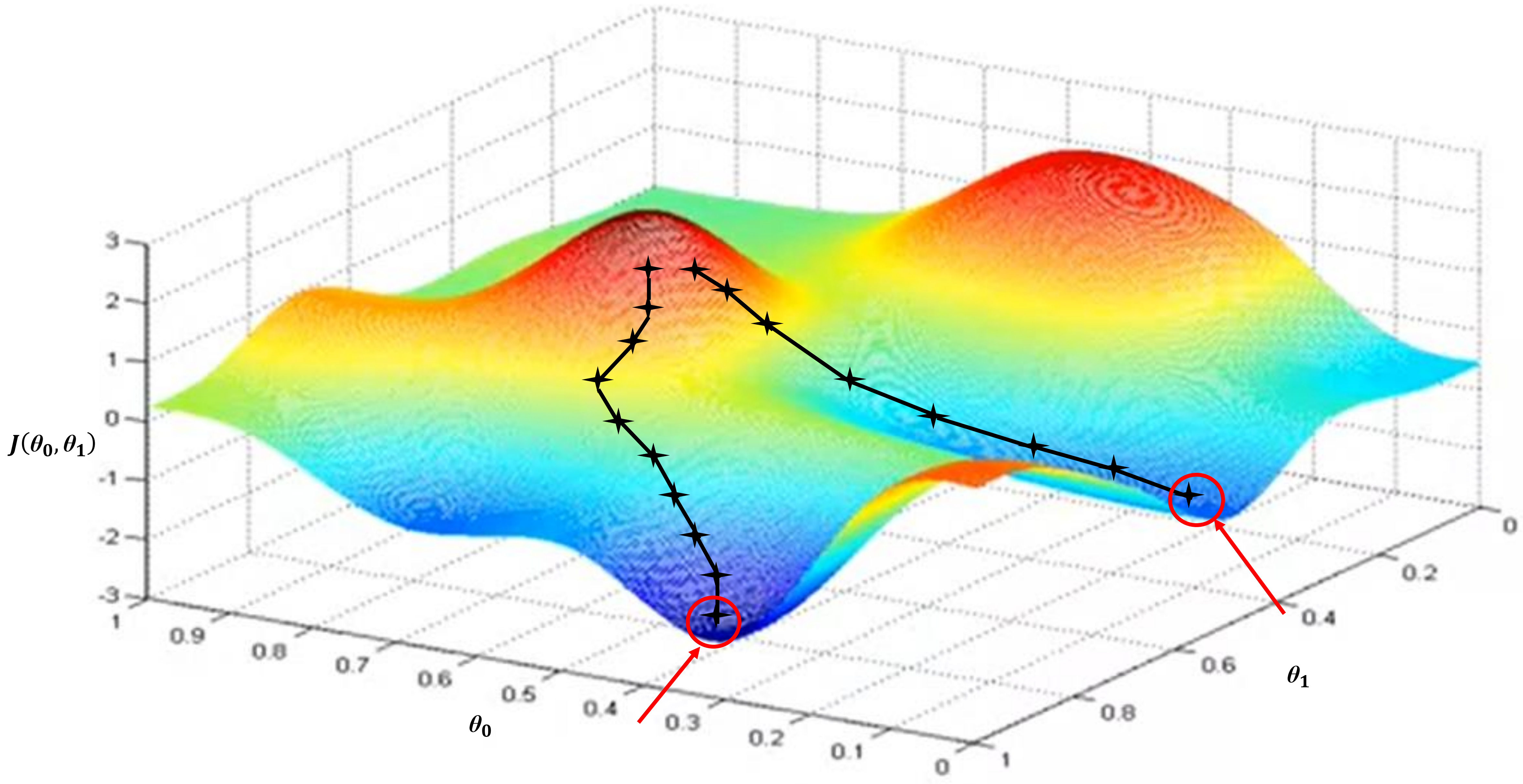
Goal: $\underset{\theta_0, \theta_1}{\text{minimize}} \; J(\theta_0, \theta_1)$

Outline:

- Start with some (θ_0, θ_1) $\theta_0 = 0, \theta_1 = 0$

- Keep changing θ_0, θ_1 to reduce $J(\theta_0, \theta_1)$ until we hopefully end up at a minimum.

Source: Machine learning course (Andrew Ng)



Gradient descent algorithm

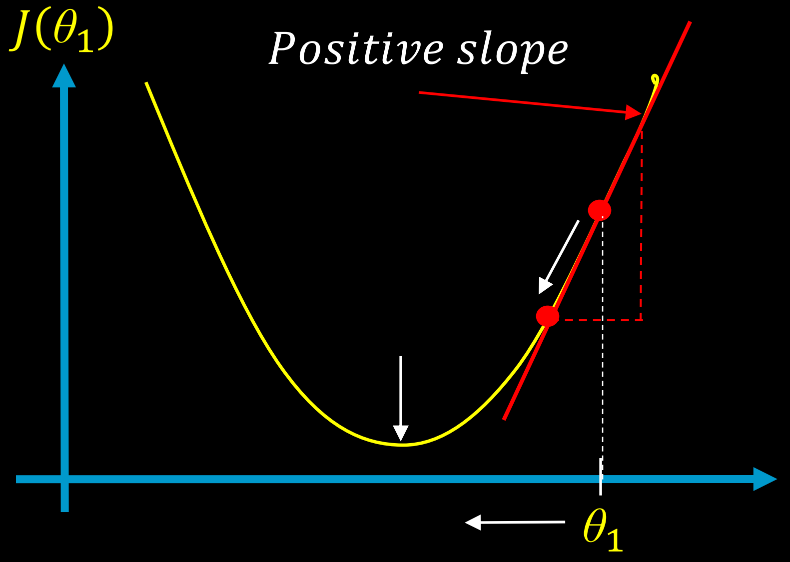
Repeat until convergence {

$$\theta_j = \theta_j - \boxed{\alpha} \cdot \boxed{\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)} \qquad (\text{for } j = 0 \text{ and } j = 1)$$

}

Learning rate **Derivative part**

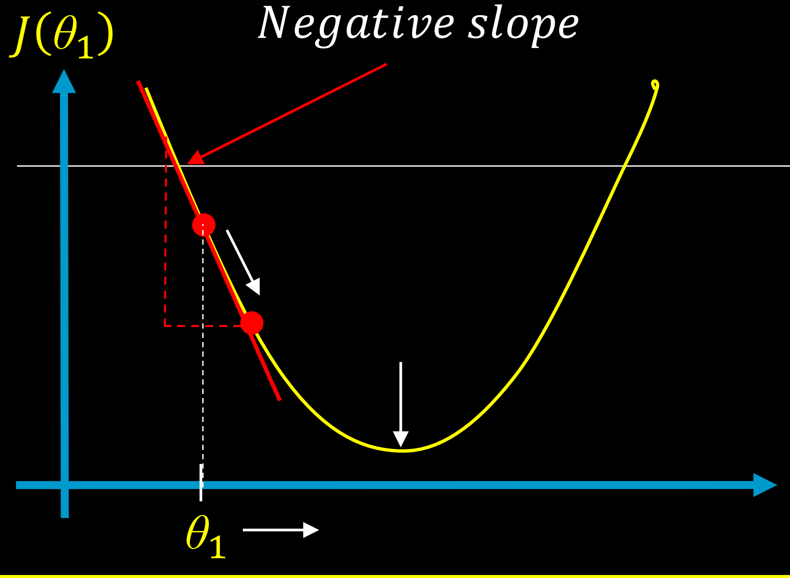
$$\underset{\theta_1}{\text{minimize}} \; J(\theta_1) \qquad \theta_1 \in R$$



$$\theta_1 = \theta_1 - \alpha \cdot \boxed{\frac{\partial}{\partial \theta_1} J(\theta_1)}$$

$$\geq 0$$

$$\theta_1 = \theta_1 - \alpha \cdot (Positive\ value)$$



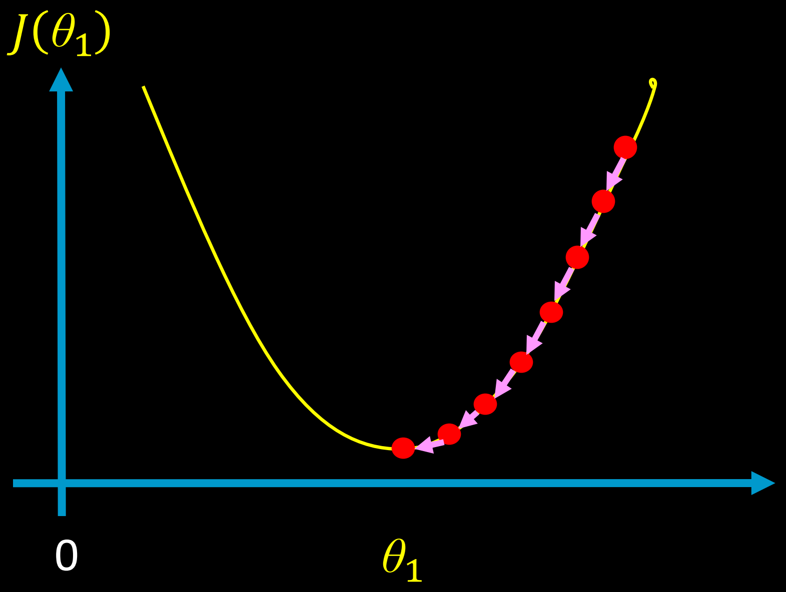
$$\theta_1 = \theta_1 - \alpha \cdot \boxed{\frac{\partial}{\partial \theta_1} J(\theta_1)}$$

$$\leq 0$$

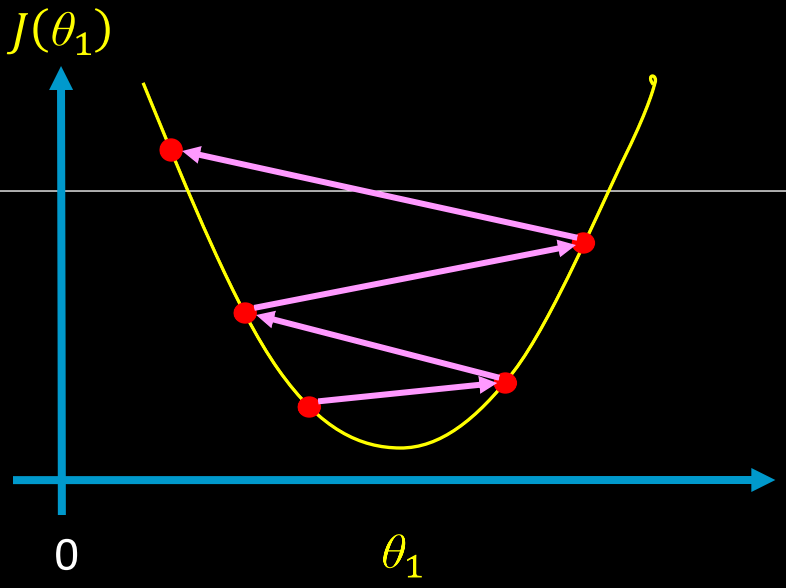
$$\theta_1 = \theta_1 - \alpha \cdot (Negative\ value)$$

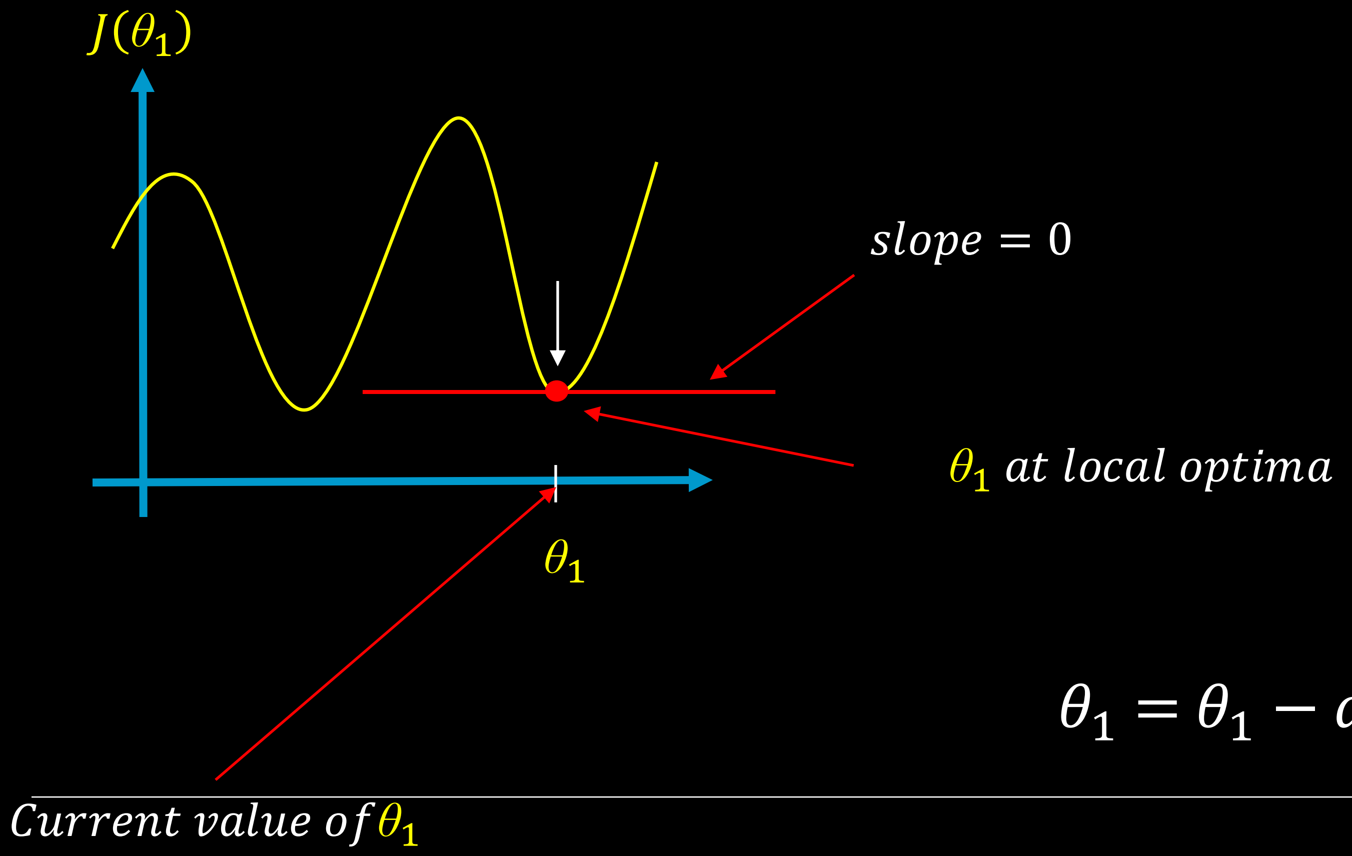
$$\theta_1 = \theta_1 - \alpha \cdot \frac{\partial}{\partial \theta_1} J(\theta_1)$$

If α is too small, gradient descent *can be slow.*



If α is too large, gradient descent can overshoot the minimum. It may fail to converge, or even diverge.





$$\theta_1 = \theta_1 - \alpha \cdot \underbrace{\boxed{\frac{\partial}{\partial \theta_1} J(\theta_1)}}_{= 0}$$

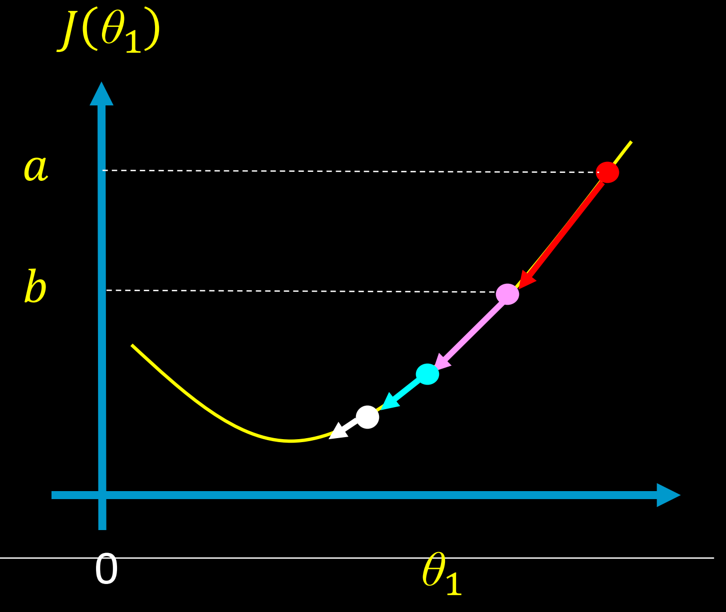
$$\theta_1 = \theta_1 - \alpha \cdot 0$$

$$\theta_1 = \theta_1$$

Gradient descent can converge to a local minimum, even with the learning rate α fixed.

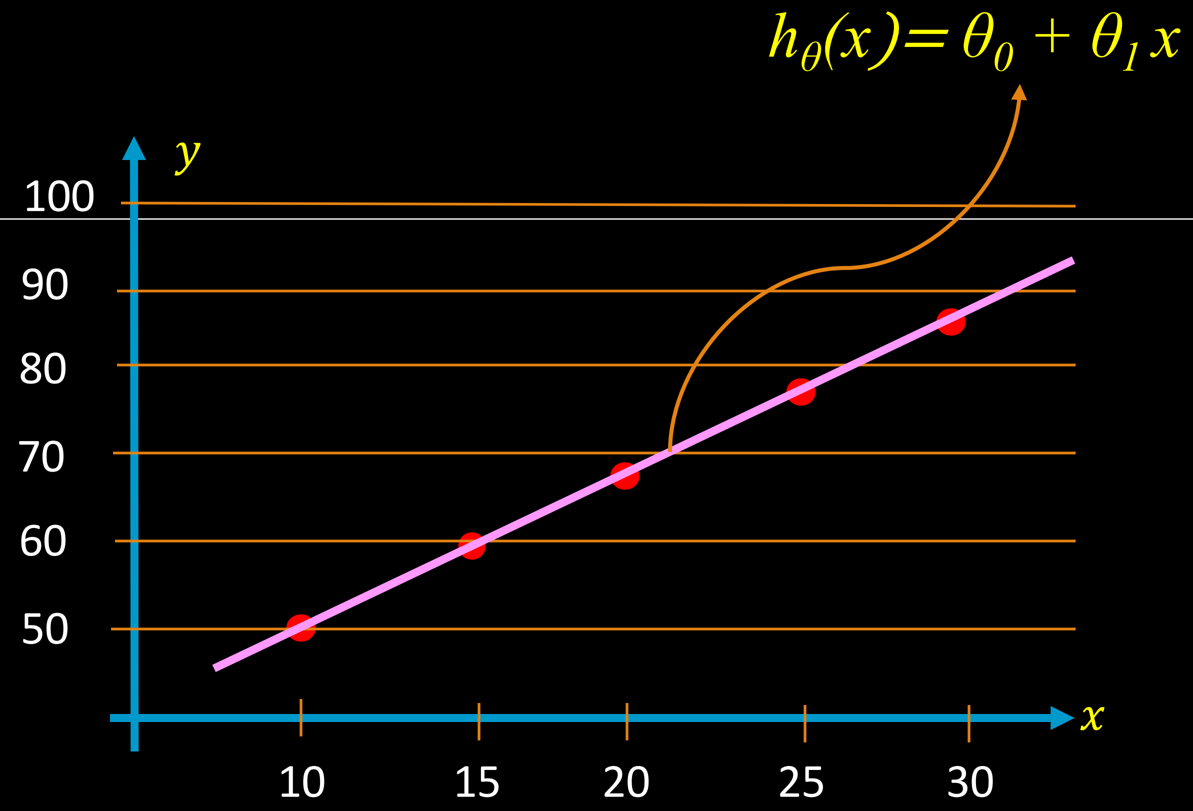
$$\theta_1 = \theta_1 - \alpha \cdot \boxed{\frac{\partial}{\partial \theta_1} J(\theta_1)}$$

As we approach a local minimum, gradient descent will automatically take a smaller steps. So, no need to decrease α over time.



Temperature
(Dataset)

Temp (°C) (x)	Temp (F) (y)
10	50
15	59
20	68
25	77
30	86



F = 32 + 1.8·t

$$h_\theta(x) = 32 + 1.8 \cdot x$$

$\theta_0 = 32$

$\theta_1 = 1.8$

Gradient descent algorithm

Repeat until convergence {

$$\theta_j = \theta_j - \alpha \cdot \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

(for $j = 0$ and $j = 1$)

}

Linear Regression Model

$$h_\theta(x) = \theta_0 + \theta_1 x$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

$$\underset{\theta_0, \theta_1}{minimize} \quad J(\theta_0, \theta_1)$$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_j} \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2 = \frac{\partial}{\partial \theta_j} \frac{1}{2m} \sum_{i=1}^{m} \left(\theta_0 + \theta_1 x^{(i)} - y^{(i)}\right)^2$$

$$j = 0 \Rightarrow \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right) \Longrightarrow \theta_0$$

$$j = 1 \Rightarrow \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right) \cdot x^{(i)} \Longrightarrow \theta_1$$

Gradient descent algorithm

Repeat until convergence {

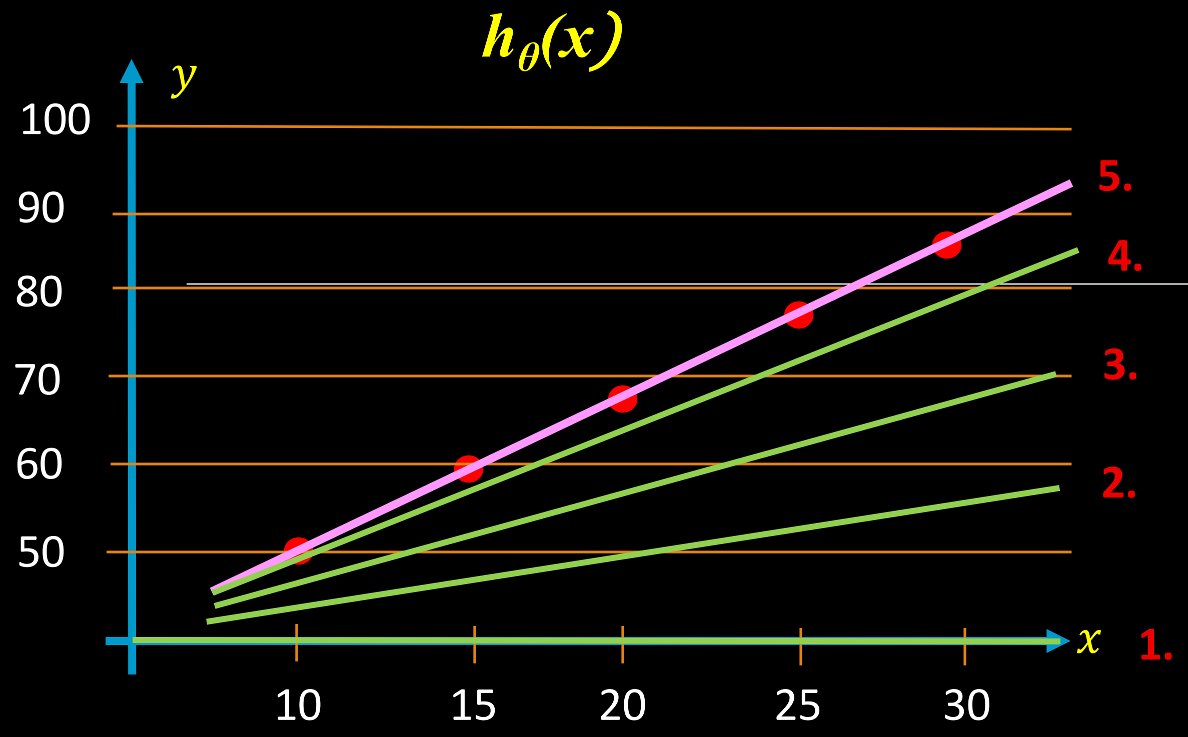
$$\theta_0 = \theta_0 - \alpha \cdot \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) \; - \; y^{(i)}\right)$$

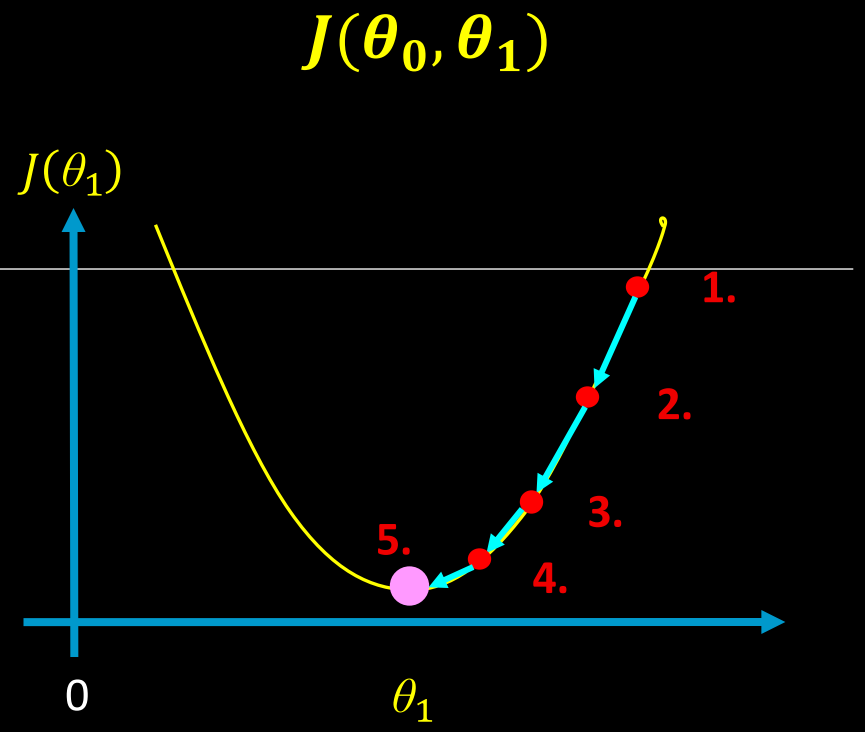
$$\theta_1 = \theta_1 - \alpha \cdot \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) \; - \; y^{(i)}\right) \cdot x^{(i)}$$

Update
θ_0 and θ_1

(for $j = 0$ and $j = 1$)

}





Iterations:

1. $\theta_0 = 0, \theta_1 = 0$
2. $\theta_0 = \ldots, \theta_1 = \ldots$
3. $\theta_0 = \ldots, \theta_1 = \ldots$
4. $\theta_0 = \ldots, \theta_1 = \ldots$
5. $\theta_0 = 32, \theta_1 = 1.8$

Cost function:

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

Gradient descent:

$$\left.\begin{aligned} \theta_0 &= \theta_0 - \alpha \cdot \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right) \\ \theta_1 &= \theta_1 - \alpha \cdot \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right) \cdot x^{(i)} \end{aligned}\right\}$$

Update θ_0 and θ_1


THANK YOU